

14.2 Finding an Unknown Value in an Equation

Lesson Introduction

 Benchmark	MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.		 Learning Targets	- I can determine an unknown decimal in a numeric equation with three numbers. - I can determine an unknown fraction in a numeric equation with three numbers.
 Benchmark Coherence	Building On MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		Working Toward N/A	
 Essential Questions	- How can you represent unknown values using variables? - How can you use reasoning to determine the value of an unknown number in an equation? - How can you use a bar model to determine the value of an unknown number in an equation? - How can you use a diagram to determine the value of an unknown number in an equation? - Why do different equations have the same solution?			
 Lesson Narrative	This lesson builds on students’ understanding to solve one-step equations with fractions and decimals using any of the four operations. The equations relate three numbers, with the unknown on either side of the equal sign in any position. The lesson emphasizes using reasoning, diagrams, and mental math to figure out the unknown instead of using inverse operations. All terms and solutions will be positive, rational numbers.			
 Component Summary	14.2.1 Warm-Up (~5 min.)	Students reflect on different careers in STEM (<i>SE p. 342</i>)	14.2.4 Collaboration (10 min.)	Students discuss how to solve equations using reasoning and diagrams (<i>SE p. 345</i>)
	14.2.2 Refresher (5 min.)	Review identifying variables and algebraic equations (<i>SE p. 342</i>)	14.2.5 Your Turn! (10 min.)	Practice figuring out the unknown in equations (<i>SE p. 346</i>)
	14.2.3 Guided Instruction (15-20 min.)	Solve equations using reasoning and diagrams (<i>SE p. 343</i>)	14.2.6 Wrap-Up (~5 min.)	Students reflect on what they learned and what they still have questions about (<i>SE p. 347</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Variable - Equation - Algebraic equation - Solve - Solution <u>Additional Terms:</u> N/A		- Career profile video (Optional) 10 × 10 grids (Optional) Fraction tiles	
		Additional Preparation 14.2.1 Warm-Up contains a video to accompany the question prompts.		

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may think they should use the operation and numbers presented in the equation rather than thinking about finding the value of the unknown. - Students may perform the calculations incorrectly if using mental math when the values do not have the same number of decimal places. - Students may believe coefficients are added to the variable rather than the multiplicative relationship of coefficient and variable. - Students may add or subtract across numerators and denominators rather than finding a common denominator or thinking about the values represented.	- Encourage students to draw their own diagrams to model the problems if they have trouble identifying with or understanding the diagrams in the lesson. - Encourage students to see if their solutions make sense in the original problem before substituting to check. Then encourage students to substitute their solutions back into the original equation to check. - Have templates of 10×10 grids available if students need to use them. - This lesson is an opportunity to further build students fluency with the operations of rational numbers. Students should not have access to calculators for this lesson. - Students should not be formally using properties of equality to solve the equations. The lesson is meant to encourage algebraic reasoning through models. - If time is an issue, then assign 14.2.4 Collaboration and/or 14.2.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk Consider utilizing Algebra Talk with the problems in 14.2.3 Guided Instruction. After presenting a problem, give students time to formulate strategies to figuring out the unknown in the equation. Ask different students how they would approach a problem. Then discuss the strategies as a group and decide how to proceed.	MA.K12.MTR.2.1: Multiple Representations Students will be working to figure out an unknown variable in multiple ways. Clarifications within this standard emphasize building understanding through modeling. Utilize the progression of the lesson to highlight the process by which students are moving from equation to diagram to solution.
 Differentiate Instruction	Consider the following strategies for additional entry points for all learners: - If play money is available, struggling students might find it helpful to use the money to help solve money-related problems. Additionally, coins like dimes, nickels, quarters, and pennies might help students make better sense of decimals. Students may not initially make sense of tenths, but they can articulate the worth of a dime compared to a dollar. 10×10 grids may be helpful for struggling students. Have blank 10×10 grids available for use. 	
Lesson Notes:		

14.2.1 Warm-Up: What Is STEM?

Component Overview: Play the Career Profile video on STEM for the entire class. After students watch the video (3:06 minutes), prompt them to reflect on the profession by answering the two questions independently.



14.2.1 Warm-Up: What is STEM?

Science, Technology, Engineering, and Mathematics (STEM) is a way of thinking and doing. STEM skills are useful to almost every career.

An individual does not need to be a master in all four areas of STEM to be successful; teamwork plays an important role. Playing to a person's strengths and working as a team creates a greater impact than working alone.

1. Describe a time where it was better for you to work as a team than alone.

Answers will vary.

2. List two jobs that involve working as a team but are in different fields of STEM.

Answers will vary. Engineers and software designers both work in teams on projects.



Consider strategies for highlighting STEM in your classroom and inspiring students for STEM work.

- Do you know someone who works in a STEM field? Have them visit or virtually meet to answer questions about their job.
- Reach out to a local STEM company and ask if they do school or community work. Most companies would love to partner with a classroom.
- Consider having students return to this lesson later in the year for project-based learning around the importance of teamwork in the workplace.

14.2.2 Refresher: The Unknown Value

Component Overview: Students will review variables and their role in algebraic expressions and equations. Remind students that any letter can be used in any situation.



14.2.2 Refresher: The Unknown Value

With your partner, complete the following.

In math, **variables** are used to represent unknown values.

In an algebraic expression, a **variable** is a letter that represents a number.

1. Circle the variable(s) in each expression.

$$\textcircled{m} + 5 \qquad 2\textcircled{x} + \textcircled{y} \qquad \frac{\textcircled{a}}{4}$$

Variables are found in algebraic equations.



An **equation** is a mathematical relation statement where two equivalent expressions or values are separated by an equal sign.

An **algebraic equation** is an equation with at least one variable.

2. Circle the algebraic equations.

$$6 + 8 = 14 \qquad \textcircled{5y = 35} \qquad \textcircled{2w + 7 = 13} \qquad \frac{12}{3} + 8 = 12$$



To **solve** an equation means to find the solution.

The **solution(s)** to an algebraic equation is(are) the value(s) of the variable(s) that makes the equation true.



Consider reviewing variables, equations, and expressions with students to activate prior knowledge. Remind students that an equation contains an equal sign, whereas an expression does not have an equal sign. Write examples of equations and expressions on the board and ask students to sort them accordingly.



Prompt students to activate prior knowledge of the definition of equation, which was taught in the previous unit. This is the first time students will formalize the definitions for variable and algebraic equations. After defining as a class, provide examples and nonexamples for students to consider. Position students to deepen understanding and address misconceptions through questions like:

- What is the role of a variable in an expression or an equation?
- Could any letter be used as the variable?
- How is an empty box similar to a variable?



14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

Component Overview: Students will begin exploring how to use reasoning and visual aids to solve algebraic equations. Tell students that they can use multiple methods to solve an equation, including logic, bar diagrams, grids, and other visual aids. Remind students that they are solving for the unknown number to make an equation true.



14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams

For some equations, the solution can be found mentally or by using reasoning.

1. The solution to the equation $n - 12.1 = 35.3$ can be found using reasoning.
 - a. Discuss with your partner how to find the value of n in the equation $n - 12.1 = 35.3$.
Answers will vary. Add 12.1 and 35.3. Substitute values for n until you find one that makes the equation true.
 - b. The equation is rewritten vertically with a box in the place of the variable. Write the solution in the box and check whether it makes the equation true.

$$\begin{array}{r} \boxed{47.4} \\ - 12.1 \\ \hline 35.3 \end{array}$$

Diagrams can also be helpful to find a solution.

2. Olivia gave equal amounts of money to each of her 3 children. Olivia gave each child \$16.75. The bar model represents the equation $\frac{m}{3} = 16.75$, where m is the total amount of money Olivia gave all of her children.

16.75	16.75	16.75
m		

- a. How can the bar model be used to find the total amount of money Olivia gave all of her children, m ?
Add $16.75 + 16.75 + 16.75$ or multiply 16.75×3 to find m , the total amount of money.
- b. What number can be divided by 3 to get a result of 16.75?

$$\frac{\boxed{50.25}}{3} = 16.75$$



Consider allowing students to create a diagram of the equation on a 10×10 grid. Encourage students to explain how their diagram relates to the equation.

Use correct vocabulary when reading the decimal numbers and hold students responsible for doing so as well. If needed, review decimal terminology and tenths with students.



Consider using the following prompts to encourage students to reason through the problem:

- How do you think you could determine the value of the variable?
- Why is it helpful to substitute the value back into the equation?

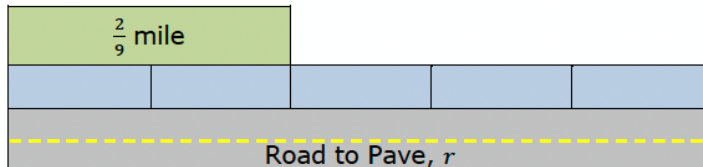


Algebra Talk

Consider using Algebra Talk as a way to encourage students to approach problem-solving in a variety of ways. Ask different students how they think the problem could be solved. Then, as a class, discuss the options presented and decide together how to proceed.

14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams (continued)

3. A road construction crew paved $\frac{2}{9}$ mile, which is $\frac{2}{5}$ of a full paving job. The bar model represents the equation $\frac{2}{9} = \frac{2}{5}r$, where r is the length of the whole road being paved.



- a. Discuss with your partner how the diagram represents the information given.

Answers will vary. The bottom and middle bars represent the entire amount to be paved. The top bar shows the amount that has already been paved.

- b. Describe how to use the bar model to solve the equation for r , the length of the road being paved.

Answers will vary. Use the top bar to figure out the length of the unpaved section and then add the amounts together.

- c. How long is the road being paved?
 $\frac{5}{9}$ mile

- d. Check your solution. $\frac{2}{9} = \frac{2}{5} \left(\frac{5}{9} \right)$



Encourage students to verbally explain the problem. Have students tell a partner what information is given and what information they need to find. Discuss how to use the bar model and if another approach may also work to solve the problem.



Scaffold instruction through intentional questioning.
 $\frac{2}{9}$ mile is $\frac{2}{5}$ of the entire paving job.

- Why do we use multiplication, $\frac{2}{5}r$, in the equation?
- How can we use the drawing to help us find the solution?
- Does your answer make sense?
- How can you check your solution?

14.2.3 Guided Instruction: Solving Equations Using Reasoning and Diagrams (continued)

4. Donté used milk for 3 batches of a recipe. He used $\frac{3}{4}$ cup of milk for each batch.



The equation $m \div 3 = \frac{3}{4}$ can be used to find m , the amount of milk Donté used.

- a. Use the measuring cups shown to solve the equation $m \div 3 = \frac{3}{4}$.

$$\frac{m}{3} = \frac{3}{4}$$

$$m = (3) \frac{3}{4} = \frac{9}{4} = 2 \frac{1}{4}$$

- b. Check your solution. $2 \frac{1}{4} \div 3 = \frac{3}{4}$



For question 4a, students may solve by adding $\frac{3}{4} + \frac{3}{4} + \frac{3}{4}$ and that is fine. They do not need to use the inverse operation as they haven't learned this yet.

14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams

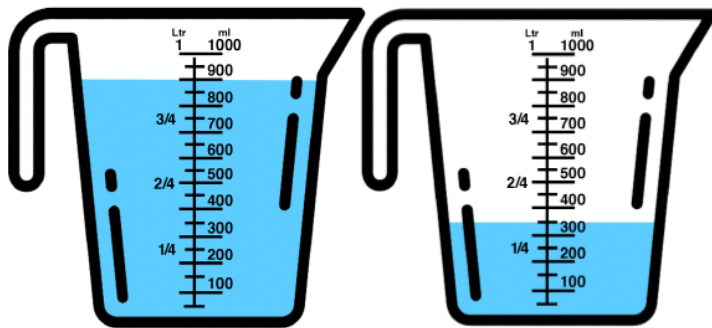
Component Overview: In the activity, students will work with a partner to solve algebraic equations by using diagrams such as measurement and bar models. Problems will use decimal numbers and fractions. This activity is a collaborative exercise, so students can either work in pairs on the same problem or work in pairs on different problems and then compare answers.



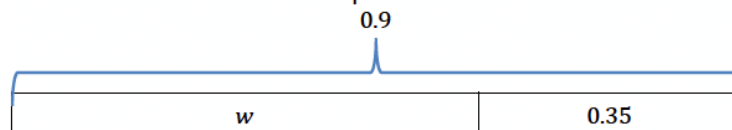
14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams

With your partner, solve the following problems using reasoning and diagrams.

- Vincent had 0.9 liters (L) of water. He poured some into the mixing bowl and had 0.35 L left.



The bar model shown represents this situation.



- Discuss with your partner how the bar model represents Vincent's situation. Then, write a brief summary of your discussion.

The entire 0.9 bar represents the total amount of water Vince had. The 0.35 portion shows the amount he poured into the mixing bowl. The bar labeled w represents the amount of water Vince has left.



Notice and Wonder

- What do you notice about the representations of the problem?
- What is the same? What is different?



If needed, review liters with students and show students an actual liter for assistance in sensemaking.

14.2.4 Collaboration: Solving Equations Using Reasoning & Diagrams (continued)

- b. The equation $0.9 - w = 0.35$ can also be used to represent this situation. Find the value of w , the amount of water that Vincent poured into the mixing bowl.

$$w = 0.55 \text{ L}$$

2. Josephine told her brother she would give him $\frac{1}{2}$ of the money they made from hosting a lemonade stand at their school carnival. She gave her brother \$24.25. This situation can be represented by the equation $\frac{1}{2}x = 24.25$ to find x , the amount of money they made at the lemonade stand.

- a. Use the bar model shown to represent the situation.

\$24.25	\$24.25
x	

- b. Find the value of x , the amount of money they made at the lemonade stand.

$$\$48.50$$



Have students make connections between the equation and the diagrams above. Encourage students to explain what each part of the equation represents and how it relates to the diagrams.

14.2.5 Your Turn!

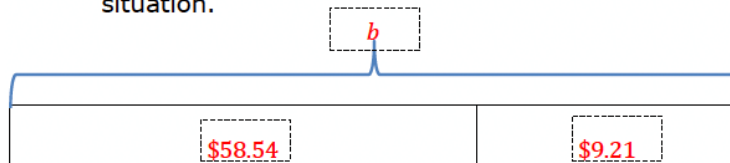
Component Overview: Students will practice solving equations with reasoning and diagrams. Each problem requires students to work with different operations. Students will work with decimal numbers when solving the problems. Remind students that they should include units in their answers, whether the units are monetary or measurement.



14.2.5 Your Turn!

1. DeAndré paid \$58.54 for a game at Store A, which was \$9.21 less than the price of the game at Store B. This situation can be represented by the equation $58.54 = b - 9.21$, where b is the price of the game at Store B.

- a. Complete the bar model diagram to represent this situation.

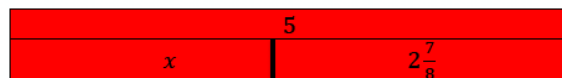


- b. Find the value of b .

$\$67.75$

2. Mr. Bing placed a new 5-gallon water bottle in the office water dispenser when he arrived in the morning. At the end of the day, the bottle had $2\frac{7}{8}$ gallons remaining. The equation $5 = x + 2\frac{7}{8}$ can be used to find x , the gallons of water dispensed from the bottle.

- a. Draw a diagram representing the situation.



- b. How much water was dispensed from the bottle? Include units.

$2\frac{1}{8}$ gallons



- How does the bar diagram help you think about how to find the solution to the given equation?
- Why do you think the “58.54” part of the bar is longer than the “9.21” part?
- Why is the answer for question 1b the sum of 58.54 and 9.21?



Students may need a quick review or explanation of what the word “dispensed” means to clear up any contextual stumbling blocks to finding the answer.

14.2.6 Wrap-Up: Reflection

Component Overview: This Wrap-Up encourages students to assess their own learning. Students will think about additional questions they have, what they have learned, and what they are still unsure of regarding solving equations by using reasoning and diagrams. Remind students to honestly answer the questions as a reflection of their learning.



14.2.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered... <i>Answers will vary but should apply to the lesson.</i>	
One mistake I learned from today... <i>Answers will vary but should apply to the lesson.</i>	One thing I am not sure about...YET! <i>Answers will vary but should apply to the lesson.</i>
One thing I am confident I know is... <i>Answers will vary but should apply to the lesson.</i>	



For students who might struggle in representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson.

- Leverage sentence stems like the Learning Targets or Essential Questions for students to reference to provide more precise responses.
- Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example.
- Mark-Up Text: Students can mark each activity with symbols that represent each of the boxes provided and explain why they chose that symbol for that activity in the corresponding box.